Geometry
Unit 10
Lesson 4 Practice

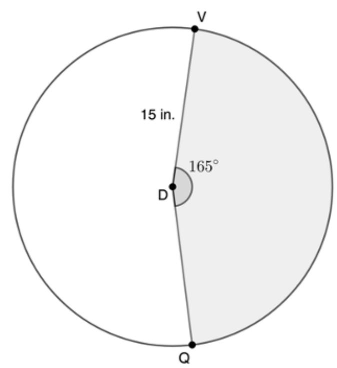
Name__Answer Key_______
Date____________________
Period__________________

1. Circle C has a diameter of 18 inches (in.). Points T and S are on the circle and form the sector TCS. The central angle $\angle TCS$ in Circle C measures $120°$. Determine the area of the sector. Express your answer in terms of π.
$A = \frac{120}{360}(\pi)(9)^2 = 27\pi$ in^2

2. The area of a sector with a radius of 5 centimeters (cm) is 15.7 square cm. Calculate the approximate measure of the central angle of the sector in degrees.
$15.7 = \frac{\theta}{360}(25)(\pi)$

$\theta \approx 71.96°$ using π button

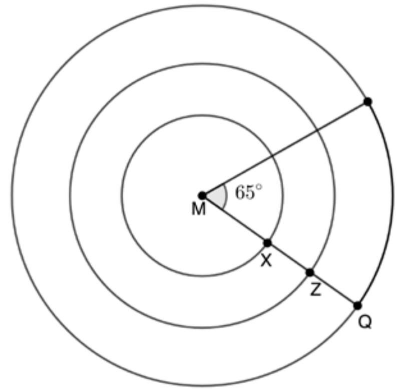
3. Circle D has radius $\overline{DV}$ that measures 15 inches (in.) with central angle $\angle VDQ$, where $m\angle VDQ = 165°$. Find the area of the shaded sector. Round your answer to the nearest hundredth.
$A = \frac{165}{360}(15)^2(\pi) \approx 323.98$ in.^{2} using π button



4. Circle K has a radius of 7 inches (in.). Points R and Y are on the circle and form the sector RKY. The central angle $\angle RKY$ in Circle K measures $145°$. Determine the area of the sector. Express your answer in terms of π.
$A = \frac{145}{360}(7)^2(\pi) \approx 19.74\pi$ in^2

5. The area of a sector with a diameter of 22 feet (ft) is 27.44 square ft. Calculate the approximate measure of the central angle of the sector in degrees.
$27.44 = \frac{\theta}{360}(11)^2(\pi)$
$\theta \approx 25.99°$ using π button

For questions 6-8, concentric circles with center M are shown with $m\angle M = 65°$. The smallest Circle M has a radius $\overline{MX}$ that measures 10 yards (yd). The largest Circle M has radius $\overline{MQ}$ and the middle Circle M has radius $\overline{MZ}$. Each circle has a radius that is 10 yd longer than the immediate prior circle.

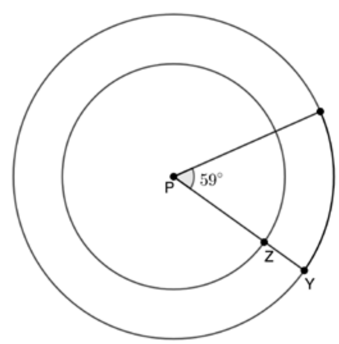


6. What is the arc length of the circle with radius $\overline{MX}$?
Arc length $= 2(\pi)(10)\left(\frac{65}{360}\right) \approx 11.34$ yds. using π button

7. What is the arc length of the circle with radius $\overline{MZ}$?
Arc length $= 2(\pi)(20)\left(\frac{65}{360}\right) \approx 22.69$ yds. using π button

8. What is the arc length of the circle with radius $\overline{MQ}$?
Arc length $= 2(\pi)(30)\left(\frac{65}{360}\right) \approx 34.03$ yds. using π button

For questions 9-10, concentric circles with center P are shown with $m\angle P = 59°$. The smallest Circle P has a radius $\overline{PZ}$ that measures 9 meters (m) and the largest Circle P has radius $\overline{PY}$. The radius of the largest Circle P has a radius 4 m larger than the smaller Circle P.



9. What is the arc length of the circle with radius $\overline{PZ}$?
Arc length $= 2(\pi)(9)\left(\frac{59}{360}\right) \approx 9.27$ m. using π button

10. What is the arc length of the circle with radius $\overline{PY}$?
Arc length $= 2(\pi)(13)\left(\frac{59}{360}\right) \approx 13.39$ m. using π button